

LYCANS AERODESIGN TEAM

Training Task 02

Avionics System Design & Battery Calculations

Deadline: Aug 22, 2026 - 11:59 PM

Submission: GitHub Repository

1 Objective

This task introduces avionics system architecture and power estimation. It mimics the workflow established in the previous assignment.

You will practice:

- Designing an avionics system architecture
- Creating a schematic in a block-diagram format
- Performing battery calculations and estimating flight time
- Writing a structured technical document using LaTeX
- Using Overleaf to compile your document
- Submitting your work through GitHub

2 Part 1 — Avionics System Design

Design an avionics system including battery calculations and a flight time estimate, with a schematic. Also, add other options you found and why did you choose your final choice.

2.1 Schematic

Create a schematic of your designed system in block-diagram format. Ensure that all components are clearly connected.

2.2 Sensors & Protocols

Include all sensors usable with the flight controller. Each sensor must be explicitly labeled with its communication protocol (UART, I2C, or SPI).

2.3 Component Selection

You must select the required components for your avionics system. Provide a table in your report matching the format below, filling in your chosen specifications for each component.

Component	Type	Reason of Choice (Optional)	Power or Amp Rating	Price	Weight
Battery					
Flight Controller					
SBC					
GPS					
Receiver					
Telemetry					
Fuse					
Fuse holder					

include additional sensors needed if not included in the flight controller

3 Part 2 — Battery Calculations & Flight Time

Based on the components in your avionics system, calculate the required battery specifications (voltage, capacity, C-rating). Provide a documented flight time estimate reflecting your power draw.

4 Report Requirements

Your research must be submitted as a minimum **two-page LaTeX document**.

Use the Lycans LaTeX template provided.

Your report must contain:

- Clear sections and subsections
- At least one relevant figure (your block-diagram schematic)
- At least one table (e.g., your component power budget)
- Properly formatted text and mathematical calculations

Do not submit screenshots of text, tables, or equations when they can be written directly in LaTeX.

5 Tools

For this task, use:

5.1 GitHub

GitHub will be used to store and submit your work.

<https://github.com/Lycans-Training-27>

5.2 Overleaf

Use Overleaf to write and compile your LaTeX document.

5.3 Diagram Software

Use any tool capable of producing clear block diagrams (e.g., draw.io, Visio, Lucidchart).

6 Submission Notes

Please ensure you follow these steps for your submission:

- **Repository:** Use the **same personal training repository** you set up previously. Do not create a new one.
- **Branching:** Create a **new branch** specifically for this task.
- **Folder Naming:** Follow the exact folder naming convention **as sent on the announcements**.
- **Pull Request:** Open a Pull Request (PR) for your submission, but **do not merge it**. The review team will handle the merge.